

Exploring the Role of the Appendage Cupping Angle on the Swimming Performance of Shrimp Through a Bio-Inspired Model

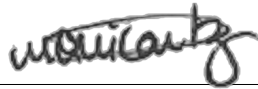
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By signing below, I attest that the undergraduate thesis listed above meets the criteria for
Honors.



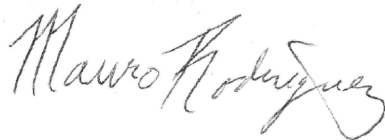
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Abstract

Through millions of years of evolution, shrimp have developed morphological adaptations that optimize their interactions with their environment, allowing for remarkable maneuverability and efficiency during swimming. The complex geometry of their swimming appendages (pleopods) enables particular configurations, such as cupping, whose contribution to efficient propulsion has yet to be quantified. Appendage cupping results from the intersection of the planes formed by the endopodite and exopodite, the biramous structures making up a pleopod. The cupping angle (ζ) enables the passive abduction of the exopodite during the power stroke and its adduction during the recovery stroke. Previous work recognized the potential significance of this mechanism to enhance force production during swimming. However, a comprehensive understanding of the fluid-structure interactions that control thrust and lift production is needed to explain how it is optimized for different swimming modes and hydrodynamic conditions. I used morphological measurements of marsh grass shrimp (*Palaemonetes vulgaris*) pleopods to guide the design of a modular robotic pleopod analog that served as a test platform for the investigation of the hydrodynamics of pleopod cupping. This method enabled parameter sweeps that would have been inaccessible using animal studies alone and enabled the independent study of the cupping angle. I quantified the thrust and lift forces produced at several cupping angle configurations using simultaneous force and particle image velocimetry (PIV) measurements at Reynolds numbers in the upper transitional flow regime. The results were compared to biological observations to determine whether animals like shrimp and krill favor an optimal cupping angle. By visualizing the wake of the model, I found that leading-edge vortices (LEVs) that remained attached to the exopodite are largely responsible for lift production. I also quantitatively determine how cupping angle impacts propulsive efficiency, primarily through its influence on the pleopod area exposed to the flow. Implementing the optimal cupping angle will ensure the maneuverability of future underwater metachronal robots under complex environmental conditions.

Chapter 1

Introduction

1.1 Relevance of Shrimp for Bio-Inspired Design

Ocean exploration and monitoring are fundamental to studying physical, chemical, and biological feedback systems on a global scale. Monitoring changes in temperature, salinity, flow, and biological variables is necessary for oceanographers and ecologists to model ocean systems accurately, a task increasingly relevant in the context of climate change [1]. While current remote sensing and localized sampling methods provide invaluable data that can be used in global models, there is a need to acquire these measurements at finer spatial and temporal resolutions. This is particularly applicable to tracking extreme fluctuations in ocean temperatures, acidification, and deoxygenation [2]. However, because of the vastness of the oceans and the technical limitations of current robotic underwater systems, much of the ocean remains unexplored, with over three-quarters of the seafloor still unmapped and over 90% of marine organisms yet to be discovered [3].

Even with advancements in sensor technology, collecting data in complex underwater environments (i.e., turbulent or rugose) remains challenging. Current solutions for in-situ ocean monitoring predominantly include remotely operated vehicles (ROVs) and autonomous underwater vehicles (AUVs). However, these underwater vehicles do not offer sufficient maneuverability, autonomy, and power to enable long-term and widespread monitoring [4]. High pressure and densely stratified ocean environments also constrain the vertical operational range of underwater vehicles [2]. Bio-inspired underwater robots provide new avenues to overcome these limitations. These systems aim to adopt physical characteristics of aquatic organisms whose evolutionary optimization over millions of years produced particular designs adapted to exploit the tradeoff between maneuverability, performance, and efficiency [5].

The core principle of bio-inspired design is not merely to copy nature but to inform the design of novel solutions that integrate existing technology using fundamental biological principles. This approach often yields improved performance and power usage compared with traditional approaches [6]. For aquatic bio-inspired design, this includes specific functional traits that arise as a need to cope with the fluid environment. Unsurprisingly, many underwater bio-inspired designs are modeled after fish. For example, tuna-inspired vehicles performed with higher efficiency than conventional AUV technologies [7, 8]. Similarly, manta rays, scallops, dolphins, and whales have also been

models for marine bio-inspired design [9, 10, 11, 12]. Most of the model organisms for bio-inspired design operate at high Reynolds numbers (Re), but we need novel solutions to navigate the transitional flow regime (traditionally in the range $1 < Re < 1000$) at finer spatial scales. This flow regime presents many challenges for locomotion because both viscous and inertial effects must be overcome. However, recent works have proposed krill as a suitable candidate for bioinspiration given their exceptional performance in this transitional Re during large vertical migrations and swarming [13, 14, 15]. They achieve this using metachronal propulsion, which is characterized by the beating of closely-spaced appendages in a phase-shifted, wave-like sequence [16]. Metachronal swimming is utilized by a wide range of taxa and scales, from microscopic protozoa to large lobsters [17, 18]. The overall success of this swimming mode in nature offers appealing new avenues to inform the development of robotic systems that can operate efficiently over a large range of Re .

Krill swim metachronally while engaging in long-distance vertical and horizontal swimming. Vertical swimming is characteristic of diel vertical migrations (DVM). DVM are daily vertical migrations largely controlled by light, where krill swarms swim through density-stratified ocean water between the ocean surface and subsurface layers. Most krill species are on the scale of 10^{-3} or 10^{-2} meters in size, yet krill DVM occurs over distances of several hundred meters in just a few hours [19, 20]. Despite this remarkable swimming performance, very little is known about the functional morphology and biomechanics of the metachronal appendages of shrimp-like organisms and how specific adaptations contribute to swimming efficiency. Krill are obligate pelagic organisms that must swim consistently to maintain posture in the water column [21]. While insightful studies came out of studying the representative krill species, Antarctic krill (*Euphausia superba*), the overall experimental difficulties they present (i.e. they do not thrive at temperatures $\geq 4^\circ$) limit our ability to conduct a more in-depth investigation of their functional morphology, particularly in the context of locomotion [22, 21, 23]. In this study, I used a morphologically and functionally similar but far more accessible species, marsh grass shrimp (*Palaemonetes vulgaris*) to measure morphological, biomechanical, and fluid data more reliably.

1.2 Shrimp Morphology

A comprehensive view of the functional morphology of shrimp is essential for identifying relevant biological mechanisms that are applicable to engineered solutions. Shrimp have five pleopod pairs, denoted as P1 through P5, arranged sequentially from the anterior to the posterior body. The proximal segment of the pleopod is called the protopodite, which branches into two distal ramal structures, the endopodite, and exopodite (Fig. 1.1). The exopodite is laterally located and is slightly larger than the endopodite. The rami spread apart during the power stroke, the phase of the beat cycle when shrimp generate drag-based thrust and lift [15, 21, 24]. During the recovery stroke, the pleopods collapse by adducting the exopodite medially (Fig. 1.3). Pleopods are flexible structures, whereby the rami bends under hydrodynamic loading from the oncoming flow during the recovery stroke. During the power stroke, the exopodite abducts into the free stream by rotating about the ramal joint but remains stiff to achieve maximum surface area and produce the most thrust [15, 22]. The endopodites of adjacent pleopods

in a pair are connected through the appendix interna, a small structure that prevents lateral separation of the endopodites [25].

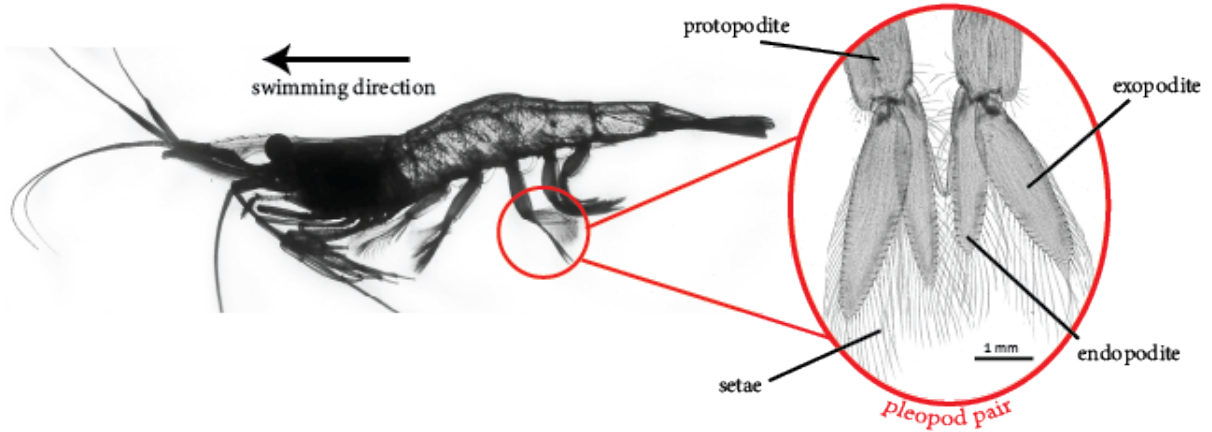


Figure 1.1: **External morphology of marsh grass shrimp (*P. vulgaris*) pleopods.** The stiff proximal segment is the protopodite, and the distal, flexible biramous segments include the endopodite and the exopodite. These rami have a solid membrane whose edges are lined with a fringe of hair-like setae. While the endopodite remains relatively fixed during a stroke, the exopodite rotates about the ramal joint to abduct in the free stream, modulating the γ angle.

Shrimp metachronal swimming kinematics is characterized by several angles (Fig. 1.2). The leg amplitude, denoted by α , is the angle between the rostrocaudal body axis (coincident with the swimming direction) and the protopodite. The β angle defines the angle between the ramal joint and the tip of each ramus. The overall angle between the body axis and the tip of the exopodite, ψ , is the compounded α and β angles characterizing the general orientation of the rami during a beat. The motions of the exopodite are defined by γ , which is its angle relative to the fixed endopodite during a beat (Fig. 1.3 (a)). During the recovery stroke, γ tends to be small, which causes the endopodite and exopodite to overlap. However, as the pleopod transitions from the recovery stroke to the power stroke, γ increases until reaching a particular maximum value (less than 90°).

The cupping angle, ζ , is the dihedral angle between the faces of the exopodite and endopodite (Fig. 1.3 (c-d)). The cupping angle lends to a “cupping behavior” exhibited by the pleopod. This is analogous to the cupping phenomenon that has been identified in fish pectoral fins and caudal fins during swimming that contribute directly to generating lift [26, 27, 28, 29]. In shrimp-like organisms, the cupping angle was first identified in *E. superba* [22], but its role during swimming and its contributions to thrust and lift production still need to be quantified.

Previous studies have provided valuable insights into single pleopod kinematics and fluid-structure interactions. Abducting the exopodite produces both lift and thrust during one beat cycle [15]. This is especially important for shrimp because they are negatively buoyant ($\rho_{shrimp} \approx 1112.78 \text{ kg/m}^3$ and $\rho_{saltwater} \approx 1030 \text{ kg/m}^3$). Thus, even during forward swimming, they must generate lift to prevent sinking. The cupping angle was suggested as a potential mechanism to enhance force production by individual pleopods towards this goal [21]. From biological observations of free-swimming *E. superba*

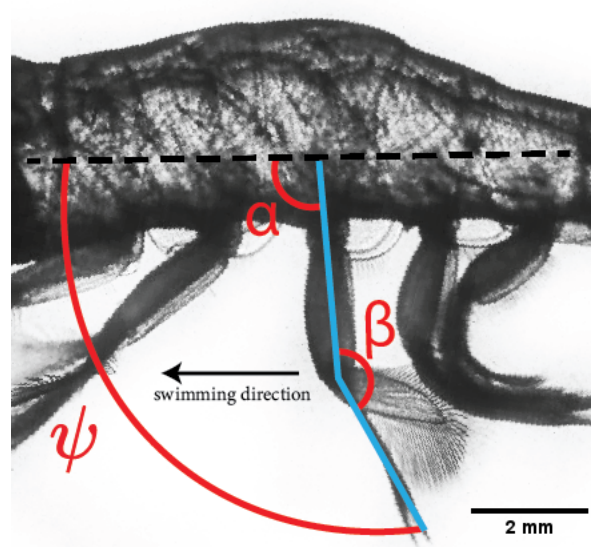


Figure 1.2: **Relevant angles that define the motions of shrimp pleopods.** α is the angle between the rostrocaudal axis of the body and protopodite. β is the angle between the ramal joint and the tip of each ramus. These two angles are used to resolve the overall motion of the leg, ψ .

and *P. vulgaris*, this angle was estimated to be between $\zeta = 35^\circ$ and $\zeta = 37^\circ$. This angle is assumed to be constant throughout a beat cycle [15]. I hypothesize that this cupping angle promotes a tradeoff between producing sufficient thrust while generating enough lift to overcome sinking and maintain posture. If the cupping angle is adjustable during shrimp swimming – which is not known from existing literature – this parameter might serve an analogous function to the angle of attack (AoA) in airfoils to modulate lift and thrust. As such, I hypothesized that the approximately 35° cupping angle seen in nature emerges as an optimal compromise to generate sufficient lift while maximizing thrust rather than strictly achieving either high lift or thrust.

1.3 Swimming Hydrodynamics

During swimming, lift is generated largely due to the production of vortices along the pleopod. Previous flow field measurements of *E. superba* have shown the formation of counter-rotating tip vortices on pleopods during swimming [21, 24, 15]. In this study, I specifically examine the leading-edge vortices (LEVs) forming along the exopodite during the power stroke. While LEVs have been reported in free-swimming *E. superba*, their origin and effects on force production remain unclear [15]. While animal studies are valuable, they often suffer from insufficient control over the parameters of interest to test biological hypotheses. Thus, I designed and operated a robotic pleopod analog that recreated the cupping angle to provide a comprehensive overview of the effects of the cupping angle while leveraging biological observations. Similar approaches have been taken to study LEV formation on the flapping wings of insects. These studies demonstrated experimentally that LEV formation induced significant lift production, especially in unsteady conditions at high angles of attack [30]. Several parameters must be considered to understand how LEV formation influences lift generation, including

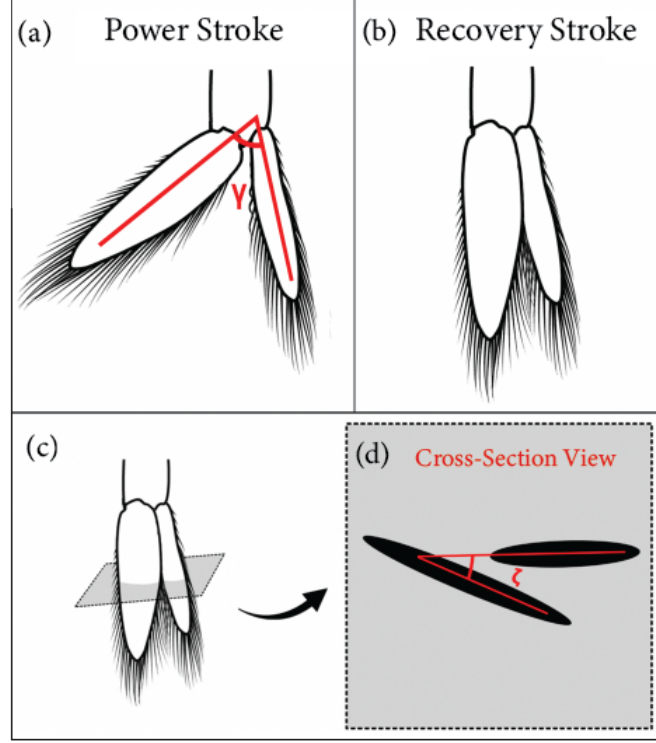


Figure 1.3: **Schematic of exopodite motion during the power and recovery strokes.** (a) The exopodite abducts during the power stroke and (b) adducts during the recovery stroke. (c-d) A cross-sectional view of the pleopod shows the cupping angle, characterized by the dihedral angle between the faces of the exopodite and endopodite.

vortex strength. Vortex strength is a measure of vorticity magnitude, which is related to circulation through the Stokes theorem:

$$\Gamma = \oint_C \bar{U} \cdot d\bar{s} = \int \int_S \bar{\nabla} \times \bar{U} \cdot d\bar{S} = \int \int_S \bar{\omega} \cdot d\bar{S} \quad (1.1)$$

where Γ is circulation, $\bar{U}$ is the fluid velocity, $\bar{\omega}$ is vorticity, and S is a surface with a closed boundary C . Furthermore, circulation can be related to lift through the Kutta-Joukowski theorem [31]:

$$F_L = -\rho_{fluid} \bar{U}_\infty \Gamma \quad (1.2)$$

where F_L is the force of lift, ρ_{fluid} is the fluid density, and $\bar{U}_\infty$ is the free-flow fluid velocity of the surrounding fluid. Based on this relationship, it is evident that more lift can be generated via stronger vortices and circulation. The attachment duration of the LEVs is also an important parameter [32]. Longer attachment duration means that the vortex does work on the propulsor for longer, thus potentially producing more lift. Using the robotic pleopod analog and a combination of prescribed cupping angles, I explore how the cupping angle influences LEV vortex strength and attachment, particularly during the pleopod power stroke. Using velocimetry and direct force measurements,

I link the formation and presence of these coherent flow structures with the forces produced throughout a beat to evaluate their role in thrust and lift.

1.4 Objectives

This investigation presents a comprehensive overview of the functional and physical mechanisms associated with the cupping angle to explore how metachronal swimmers like shrimp generate thrust and lift. Using biological data, I designed a scaled robotic pleopod analog programmed to perform real shrimp kinematics and fitted with a tunable ramal joint to modulate the cupping angle. I explored how cupping influences lift and thrust production by conducting a parameter sweep of cupping angle and Re . Three experimental methods were used concurrently to evaluate the role of the dominant flow structures on force production: image tracking of the model kinematics, particle image velocimetry (PIV), and direct force measurements to quantify the drag, thrust, and lift force components during swimming. I linked the instantaneous lift and thrust forces with the wake features of pleopod actuation. The tradeoff between lift and thrust was also evaluated by comparing the ratios of these forces across cupping angles. Lastly, I discuss how the cupping angle can be tailored to specific swimming modes like hovering, forward swimming, or vertical migrations. Future maneuverable ocean technologies will demand dynamic transitions between these movements, and the insights of bio-inspired robotics can be leveraged to optimize these modern designs.

Chapter 2

Methods

2.1 Pleopod Model Design

The mean leg morphometrics and mean α and β kinematics data of *P. vulgaris* ($n = 10$ shrimp) was used to design and operate a simplified 3D-printed robotic pleopod model (Fig. 2.1). Only the endopodite and exopodite were included in the model because the protopodite has a negligible influence on the hydrodynamics of the pleopod as a whole [21]. The model was dynamically scaled based on Re to $40\times$ larger than the pleopods of *P. vulgaris* to make the robotic pleopod easy to integrate with standard hardware. Re was defined based on the velocity and length scale of the exopodite paddle:

$$Re = \frac{U_{tip} * W}{\nu} = \frac{2\pi\theta fWL}{\nu} \quad (2.1)$$

where W is the chordwise width of the exopodite, U_{tip} is the cycle-average speed of the tip of the exopodite, and ν is the kinematic viscosity of the fluid. Mean pleopod U_{tip} during a beat cycle was calculated from f , the beat frequency of the pleopod, θ , the stroke amplitude of the pleopod, and L , the length of the exopodite from the ramal joint to the tip. The resulting Re of the model was within the biologically relevant range of around 600 to 1700, encompassing the Re of both *E. superba* and *P. vulgaris* [15, 33]. As opposed to achieving the desired Re range by only varying the pleopod speed, the viscosity of the surrounding fluid was also increased. This allowed operating the model fast enough to produce measurable forces. All experiments were performed in a glycerin-water mixture (a 2:3 glycerin:water ratio) with a dynamic viscosity of $\mu = 0.0114 \text{ kg m}^{-1} \text{ s}^{-1}$, which was measured with a rotational rheometer (Ares-G2, TA Instruments). The density, measured with a standard hydrometer (Chase Instruments, NY, USA) was $\rho = 9.8652 \times 10^{-6} \text{ kg m}^{-3}$.

The 3D model was designed in Fusion 360 (Autodesk Inc., San Francisco, CA, USA) and manufactured using resin 3D printing with an ELEGOO Saturn 2 8k resin printer (Elegoo Industries, Shenzhen, China). The two endopodite and exopodite paddles were secured at a ball-bearing joint to reduce friction as the exopodite rotates during the power and recovery strokes. Because the resin was denser than the water-glycerin mix, I made the exopodite neutrally buoyant by draining resin from an uncured region within the exopodite. The uncured region was positioned as close to the bearing joint as possible to reduce the moment on the paddle from the non-uniform mass distribution

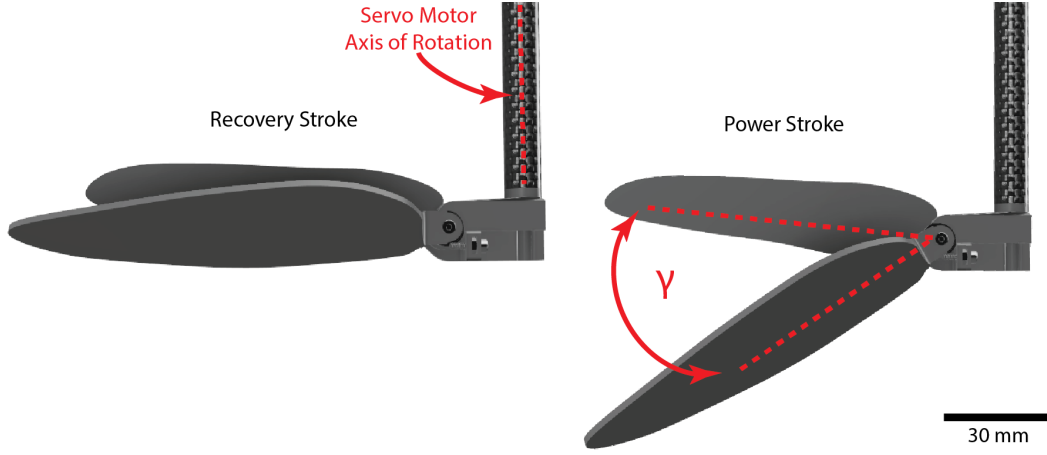


Figure 2.1: **CAD rendering of the simplified 40 $\times$ robotic pleopod model.** The geometry of the endopodite and exopodite is representative of *P. vulgaris*. The pleopod was oriented horizontally (at 90° relative to their natural orientation) to keep the driving servo above the waterline while ensuring the pleopod was fully submerged. This resulted in the exopodite extending downwards instead of sideways. The model was made neutrally buoyant to eliminate gravitational and buoyancy effects.

after draining resin. Refraction of light through the exopodite during PIV was avoided by coating the model with a thin layer of matte black spray paint.

Small swappable exopodite joints were printed at relevant angles to adjust the model’s cupping angle. The maximum γ angle could also be changed through a continuous range of 0° through 80° by mechanically limiting the opening of the exopodite with an adjustable bolt in the endopodite. This mechanical limit was inspired by a physical γ -limiting joint structure in *Halocaridina rubra* (see microscopy images in Appendix Fig. 5.4). The maximum γ in the model was restricted to a maximum of 70° in all experiments. This value reflects what is observed as the maximum γ for *E. superba* [22], as the maximum γ for *P. vulgaris* is not sufficiently documented. Furthermore, restricting γ based on *E. superba* provides greater consistency with a previous study incorporating the cupping angle into a krill-inspired robotic platform [15].

The rotational movement of the pleopod model to achieve the desired ψ angle was driven by a servo (HS-85BB, Hitec Commercial Solutions, San Diego, CA) attached to the vertical carbon-fiber rod holding the model (Fig. 2.1). The ψ angle was calculated from α and β as $\psi = \alpha + 180^\circ - \beta$. The kinematics of the servo recreated accurate shrimp kinematics (error = 0.1517°), determined from biological observations of *P. vulgaris* (Fig. 2.2). The biologically observed stroke cycle period was adjusted to the period defined by the model’s Re. The kinematics analysis served as an additional validation step to ensure the simplified pleopod model was sufficient in emulating real shrimp swimming.

The change in γ occurred passively. Abduction and adduction resulted from the hydrodynamic loading of the exopodite, whose orientation (i.e., abduction status and cupping angle) relative to the flow during the beat cycle induced a force orthogonal to the exopodite. These hydrodynamic conditions are further discussed in Section 3.2.

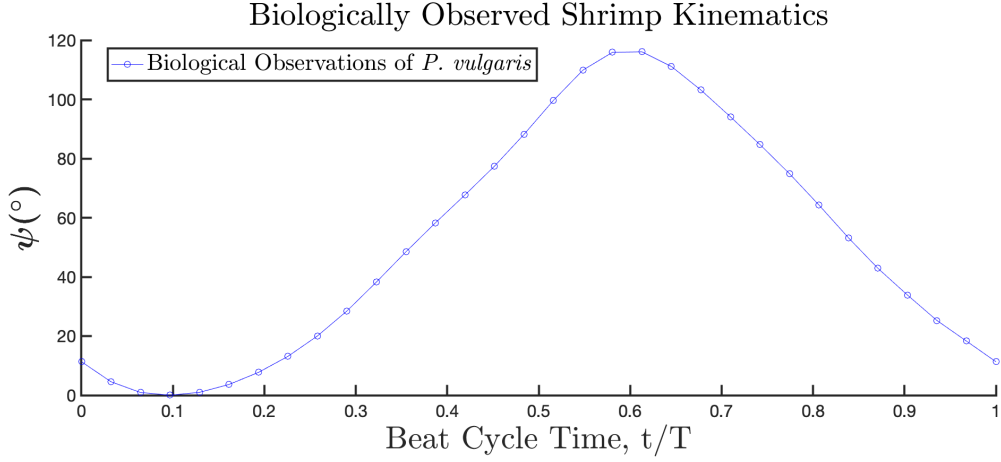


Figure 2.2: Angle ψ , over one beat cycle for pleopod P4. The ψ angle defines the pleopod kinematics as $\psi = \alpha + 180^\circ - \beta$. The time is normalized by the pleopod beat period, T .

2.2 Experiments

Experiments were conducted in a 210-liter tank ($119 \times 43 \times 41$ cm³) filled with a 2:3 glycerin-water mixture. The performance of the pleopod model was assessed while undergoing a parameter sweep of cupping angles at four different Re . Cupping angles of 0° through 80° in 10° increments were tested, in addition to a 35° angle representative of the cupping angle observed in biological studies of *E. suberba*. Re of 387, 645, 968, and 1291 were defined using [2.1](#). To run experiments at these Re , f was varied between 0.1164 Hz, 0.1940 Hz, 0.2910 Hz, and 0.3880 Hz, respectively. This was accomplished by modulating the number of kinematics steps input to the servo.

2.2.1 Kinematic Tracking

The kinematics of the pleopod model were tracked to record γ over time with different cupping angle attachments. This was done to calculate the change in surface area incident to the flow during the beat cycle. A high-speed camera (NOVA-R3, Photron, Tokyo, Japan) with a fixed focal length lens (AF-S NIKKOR 50mm, Nikon, Tokyo, Japan) was used to record the abduction and adduction of the appendage. Videos were recorded at 30 frames per second for four consecutive beat cycles. All images were processed using the DLTdv8 tracking software in MATLAB [34](#). The model was tracked at three points to measure γ : the bearing joint, a point on the exopodite, and a point on the endopodite (Fig. [2.3](#)).

The camera was positioned to record normal to the face of the endopodite when halfway through the power stroke (at approximately $t/T = 0.25$, where t/T is the time normalized by the duration of one beat cycle). The beginning of a beat cycle was identified as the frame of perceived minimum endopodite length (out of a plane in front of the focal plane), as the endopodite is fixed and its projected length in the frame changes proportionally to its in-and-out of plane motion relative to the camera's focal plane. To account for the parallax effect due to the model periodically moving

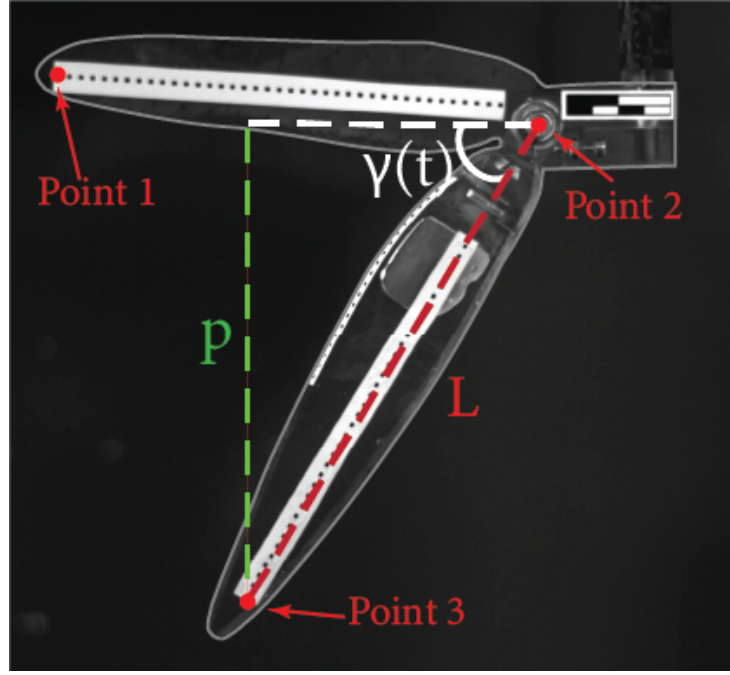


Figure 2.3: **The three tracking locations on the model used during kinematics experiments.** $\gamma(t)$ is defined by the angle between the endopodite and exopodite segments passing through points 1,2 and 3,2, respectively. The model's edges have been outlined in grey to increase contrast with the background. In the frame of reference of the model, L is the actual exopodite length, measured from Points 2 to 3. P is the projected length of the exopodite in the z -direction. These parameters determine the instantaneous γ while considering the parallax effect resulting from our camera's fixed perspective.

away from the focal plane, I calculated the γ angle in the model rather than the lab frame of reference. This was done by measuring the projected length of the actuating exopodite along the vertical lab frame axis (Fig. 2.3). This projection is common to both the lab and model frames of reference. The camera's perspective does not distort its length as the model oscillates. Because the length of the exopodite is fixed, the γ can be calculated in the model frame of reference as:

$$\gamma(t) = \arcsin\left(\frac{p}{L}\right) \quad (2.2)$$

where L is the length of the endopodite measured from the joint to the tip when the exopodite face is normal to the camera and p is the projected exopodite length when projected onto the z -axis.

To calculate the instantaneous effective surface area of the paddle to the flow, the pleopod surface area was initially captured from one reference frame in which the endopodite was facing the camera. ImageJ was used to measure the surface area of the exopodite and endopodite in these cases. Note that the surface area of the endopodite does not change between trials, but the exopodite surface area incident to the flow is reduced as the cupping angle increases and γ decreases (causing the rami to overlap). The incident area varies from a maximum when $\zeta = 0^\circ$, to a minimum when $\zeta = 80^\circ$. The instantaneous γ values calculated earlier were used to calculate the change in

surface area when the exopodite overlapped with the endopodite.

2.2.2 Flow Field Measurements

Velocity and vorticity fields were obtained using PIV. A horizontal laser sheet was used to orthogonally bisect the exopodite paddle at maximum abduction ($\gamma = 70^\circ$). Two lasers (Optotronics, 4W at 532 nm) were employed to project overlapping laser sheets, thus reducing shadows and achieving uniform illumination throughout the field of view. A coordinate system is established in Fig. 2.4 to place the experimental setup within the context of the direction of lift and thrust produced by the model. The direction of lift and thrust forces indicate the direction of forward and vertical swimming had the model been replaced by a real shrimp. The forward and vertical swimming directions, which are orthogonal, lie within a plane parallel to the bottom of the tank. The z -direction is normal to this horizontal plane. The same high-speed camera from kinematics experiments was used for PIV and recorded 250 frames per second. The camera was positioned to record the plane parallel to the bottom of the tank since the exopodite abducted downward. The resulting focal plane is in the same laser plane through which the exopodite extends during abduction (Fig. 2.4). The camera was mounted laterally to the tank, and the view was projected using a mirror oriented at 45° because of limited space under the experimental tank. The fluid was seeded with $10\ \mu\text{m}$ reflective hollow glass spheres (LaVision, Göttingen, Germany). Fluid velocity fields were computed using the DaVis 11 software package (LaVision, Göttingen, Germany). Image pairs were analyzed with three passes of overlapping interrogation windows with decreasing areas from $48\ \text{pixels} \times 48\ \text{pixels}$ (50% overlap, 3 passes) to $32\ \text{pixels} \times 32\ \text{pixels}$ (75% overlap, 3 passes). No smoothing of the velocity data was performed, but denoising was carried out using an average vector from NaN neighborhood in a 3×3 grid.

2.2.3 Force Measurements

Force measurements were performed synchronously with PIV recordings to ensure the measurements accurately matched the recorded instantaneous hydrodynamic interactions between the model and surrounding flow. Forces were measured using a six-axis force transducer (resolution = $0.003125\ \text{N}$, Nano 17 F/T transducer, ATI Technology, Toronto, Canada). The signal-to-noise ratio based on the peak-to-peak force signal induced by oscillations of the model was 22.4, confirming that the force resolution was sufficient for this application. The force transducer was mounted between the servo and pleopod model so that the force transducer rotated with the model. The transducer records forces along three directions: x , y , and z . Forces in the direction of x and y were in the same plane as the forward and vertical swimming directions, but not necessarily in the same direction. The exopodite face was aligned with the $+x$ -direction of the force transducer in all trials. The $+y$ -direction was normal to the face of the exopodite, and the model rotated about the z -axis of the force transducer. The z -axis of the force transducer is parallel to the z -direction in the lab frame of reference. Despite the rotating force transducer axes, the lift and thrust direction remained the same throughout a beat cycle in the directions established by the coordinate system in Section 2.2.2 Fig. 2.4. The force transducer was interfaced to a DAQ (NI USB-6210, National

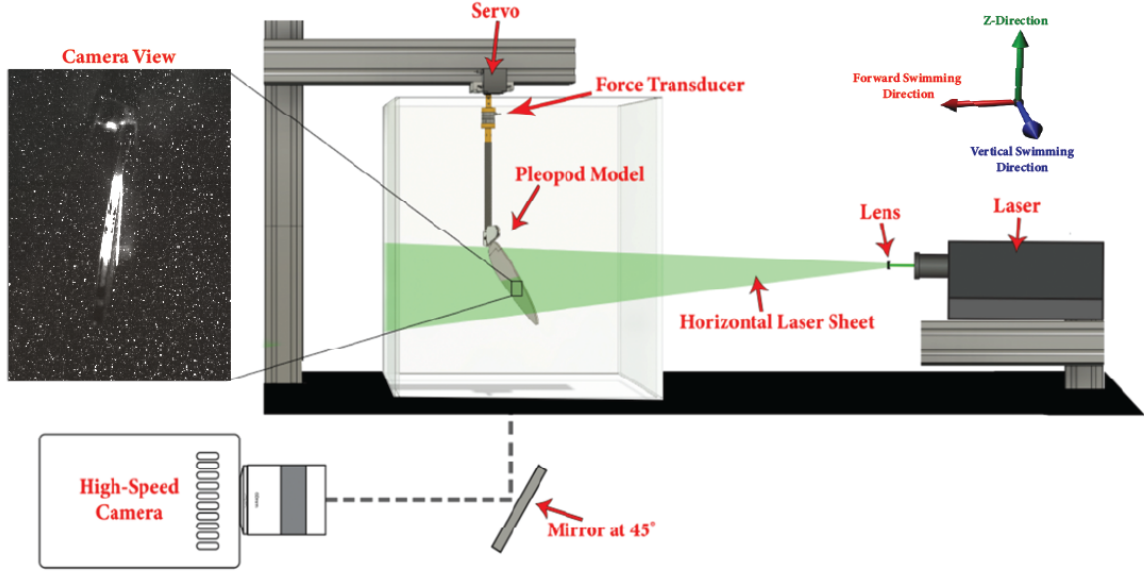


Figure 2.4: **Rendering of the experimental setup for force and PIV data collection.** The horizontal laser sheet bisects the fully abducted exopodite. The force transducer is mounted between the servo and the model using custom adapters, aligning the x -axis of the force transducer with the face of the endopodite. Note that the experimental tank was larger relative to the model than depicted in this schematic to minimize wall effect during the experiments.

Instruments, USA) to read (sampling rate = 100 Hz) and convert the voltage magnitude to force data. The force recordings for each beat cycle were split into power and recovery strokes and then averaged throughout the stroke ($n = 4$ consecutive strokes). Multiscale principal component analysis (MSPCA) was used, along with a Daubechies wavelet (Daubechies 4), to denoise the raw force signal [35, 36, 37]. Different wavelets were tested and the wavelet with the lowest Root Mean Square Error between the raw data and the filtered data was chosen. The methodology of this approach is similar to that of multiscale Kalman filtering and multi-block PCA and involves decomposing each variable based on the selected family of wavelets [35].

Because the force transducer axes rotate over time relative to the forward and vertical swimming direction in Fig. 2.4, the lift and drag-based thrust were decomposed into F_x and F_y . This was accomplished by considering the orientation of the model according to the shrimp kinematics, described by ψ , and the cupping angle, ζ (Fig. 2.5). For example, if we consider a pleopod at the beginning of the beat cycle with no cupping angle ($\psi = 0$ and $\zeta = 0$), the lift force acts normal to the face of the exopodite, hence, parallel to the y -axis of the force transducer. However, since ψ is continuous, there are infinitely many points where neither lift nor drag-based thrust acts parallel to one of the force transducer axes. I explore the lift and drag-based thrust at a discrete subset of these values by considering both ψ and ζ , which can be combined into a compound angle, $\psi - \zeta$. This angle helps describe how the forces evolve throughout a beat cycle (Eqs. 2.3 and 2.4).

$$\bar{F}_L = F_x \sin(\psi - \zeta) + F_y \cos(\psi - \zeta) \quad (2.3)$$

$$\bar{F}_T = F_x \cos(\psi - \zeta) + F_y \sin(\psi - \zeta) \quad (2.4)$$

Compound angle for all ζ

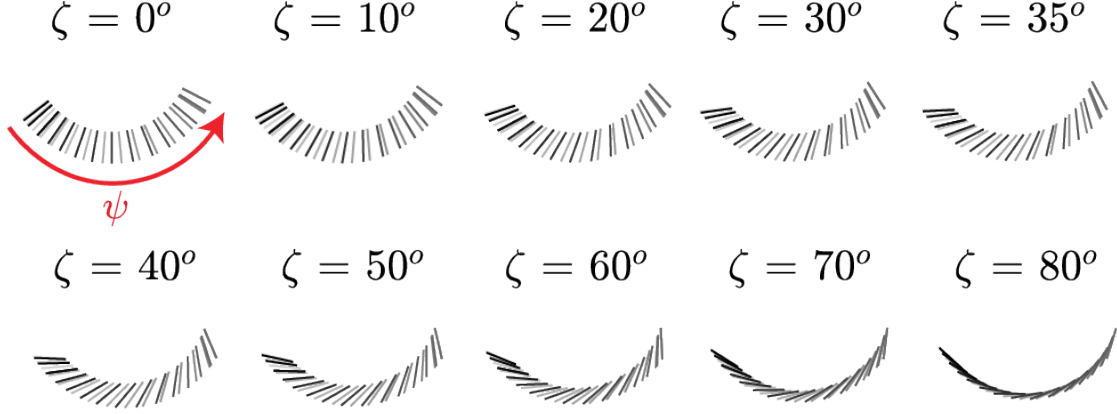


Figure 2.5: **Visualization of the orientation of the cross-section of the exopodite as the compound angle evolves with ψ and ζ .** For each cupping angle, a cross-sectional view of the exopodite (the same exopodite cross-section as in Fig. 1.3) is shown at several time steps throughout a beat cycle. The color of the exopodite transitions from black to light grey to show the progression of time from the start of the power stroke to the end of the recovery stroke, respectively.

Lift and drag coefficients, C_L and C_D , were determined based on the resulting forces (Eqs. 2.5 and 2.6):

$$C_L = \frac{2\bar{F}_L}{\rho\bar{U}^2 A} \quad (2.5)$$

$$C_D = \frac{2\bar{F}_D}{\rho\bar{U}^2 A} \quad (2.6)$$

The average force over the power and recovery strokes was calculated. The standard deviation for each point was calculated using the variance formula for error propagation (Eq. 2.8).

$$\sigma_{F_L}^2 = \left(\frac{dF_L}{dF_x}\right)^2 \sigma_{F_x}^2 + \left(\frac{dF_L}{dF_y}\right)^2 \sigma_{F_y}^2 + \left(\frac{dF_L}{d(\psi - \zeta)}\right)^2 \sigma_{(\psi - \zeta)}^2 \quad (2.7)$$

Similarly, for thrust

$$\sigma_{F_T}^2 = \left(\frac{dF_T}{dF_x}\right)^2 \sigma_{F_x}^2 + \left(\frac{dF_T}{dF_y}\right)^2 \sigma_{F_y}^2 + \left(\frac{dF_T}{d(\psi - \zeta)}\right)^2 \sigma_{(\psi - \zeta)}^2 \quad (2.8)$$

where σ_{F_L} , σ_{F_T} , σ_{F_x} , σ_{F_y} , and $\sigma_{(\psi - \zeta)}$ is the standard deviation for the lift force, drag-based thrust force, transducer force in the x -direction, transducer force in the y -direction, and compound angle, respectively.

Chapter 3

Results and Discussion

3.1 Kinematics

Overall, the model produced kinematic trends that were relatively consistent with biological kinematic data (Fig. 3.1). No data exists in the literature confirming whether exopodite actuation in shrimp is a passive (i.e., caused by fluid-structure interactions) or an active process (i.e., muscle-powered) or report the instantaneous γ during a pleopod beat cycle. As such, I used reference data from unpublished γ kinematics of *P. paludosus* to evaluate the motions of the robotic exopodite. The commonalities between biological and robotics trends suggest that the γ angle is largely driven by the passive effects of hydrodynamic loading of the hinged exopodite during the power and recovery strokes of the pleopod. Still, it must be noted that the exopodite adducts much earlier in *P. paludosus* during the final phase of the power stroke (rather than during the recovery stroke) when the velocity of the appendage decreases as seen with the decreasing γ angle before $t/T = 0.5$ (Fig. 3.1). While this could result from active actuation, it is likely that the muscle-tendon groups in the pleopods act as a spring-damper system to facilitate adduction [38].

The robotic data differs from the biological observations in some cases by showing two local maxima at peak γ amplitude. This is an artifact of the physical γ -limiting mechanism, against which, for certain cupping angles, the exopodite bounced due to excessive contact (Fig. 3.1). The muscle-tendon groups within shrimp pleopods likely mitigate these effects [38]. However, this collision plays a minimal role in the resulting kinematics and forces. Rather, this phenomenon served as a tool to more easily identify which cupping angles caused maximum abduction of the exopodite. It is important to note that the maximum γ of the robot is not expected to perfectly match the biological data acquired with *P. paludosus*. While I aimed to recreate the general motions of the pleopods in space and time (given no comparative data exists in the literature), I chose the maximum γ of *E. superba* as a reference for consistency with previous studies [15].

The kinematic analyses show that moderate cupping angles – specifically $20^\circ \leq \zeta \leq 40^\circ$ – resulted in the earliest and fastest exopodite abduction and adduction. Note that the γ limit of 70° was not reached for all cupping angles. $\zeta = 70^\circ$ and $\zeta = 80^\circ$ have maximum γ values significantly below $\gamma = 70^\circ$, only abducting to $\gamma = 44^\circ$ and $\gamma = 40^\circ$, respectively. The fastest exopodite abduction was observed for a cupping angle of 40° (Fig. 3.1). The exopodite abducted at a mean speed of 308.35° per cycle period in this

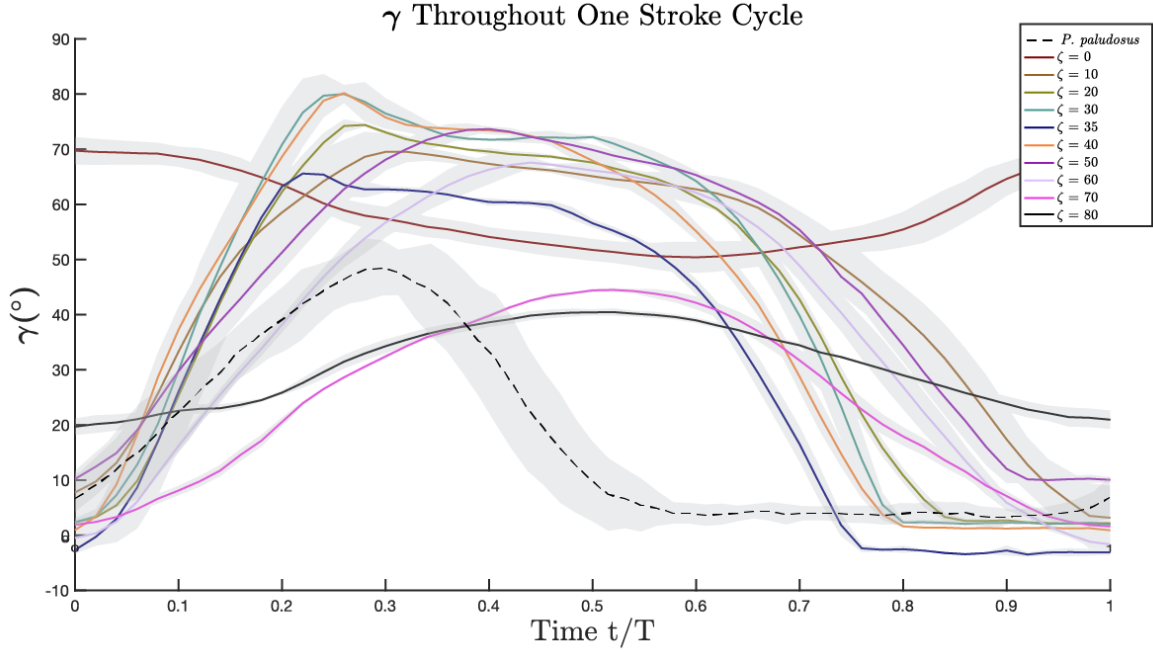


Figure 3.1: **Robotic exopodite abduction angle of one pleopod (P4), γ , throughout one beat cycle at $\text{Re} = 968$.** Time is nondimensionalized by the beat period, T . Shading indicates the standard deviation from the plotted mean values (solid lines) of 3 consecutive beats. The black dashed line represents biological observations obtained from the kinematics of tethered *P. paludosus*. This biological data is used to qualitatively validate that the model can effectively recreate real shrimp kinematics. Data was averaged for one leg (P4) over three consecutive beats.

case. Similarly, the $\zeta = 40^\circ$ exopodite fully adducted the fastest as the recovery stroke began. Cupping angles of 20° , 30° , and 35° achieved the maximum γ limit comparably fast within the beat cycle at speeds of 265.69° , 307.69° , and 275.00° per cycle period, respectively. This shows moderate cupping angles promote swift transitions between the power and recovery stroke. However, to contextualize the effect of thrust maximization and drag reduction, we must examine the extent to which the surface area of the pleopod is modulated by γ .

As expected, the robotic pleopod passively maximizes its area during the power stroke for all cupping angles (Fig. 3.2). Conversely, the area is minimized in the recovery stroke. It is important to note that surface area reduction occurs via two processes. Firstly, changes in the cupping angle directly modulate the exopodite surface area incident to the flow (i.e., larger ζ make the exopodite face less normal to the flow). Additionally, the exopodite overlaps with the endopodite during the recovery stroke, further reducing the total surface area exposed to the flow compared with the combined surface area of both paddles during full abduction. Based on Eq. 2.5 and 2.6 modulating the surface area is expected to contribute to thrust during the power stroke, followed by drag reduction during the recovery stroke (see Section 3.2).

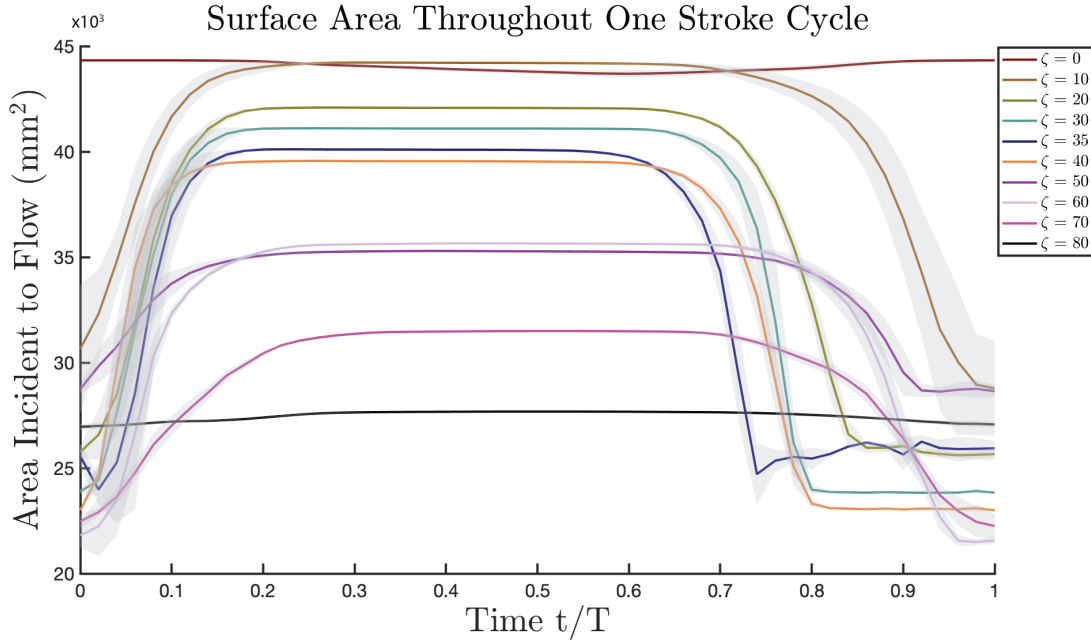


Figure 3.2: **Modulation of the surface area of the robotic pleopod exposed to the flow during one beat cycle at $Re = 968$.** Time is nondimensionalized by the beat period, T . Shading indicates one standard deviation from the plotted mean values (solid lines, $n = 3$ consecutive beats). One factor that causes changes in surface area is the cupping angle, as lesser ζ values make the exopodite face more normal to the flow. Additionally, the exopodite overlaps with the endopodite during the recovery stroke, further reducing the effective frontal area.

3.2 Lift, Drag, and Thrust Production

During one beat cycle, the generated lift exhibits local peaks and troughs that align well with particular features of the kinematic results. In the power stroke, maximum lift is generated approximately halfway through the stroke, when $t/T \approx 0.25$ (Fig. 3.4). Drawing from the kinematics findings, wherein γ was maximized at $t/T = 0.25$, the peak lift corresponds to the exopodite's full abduction (Fig. 3.1). A trough in the lift force was observed at $t/T \approx 0.45$ (Fig. 3.4). This coincided with the approximate midpoint time between full exopodite abduction and when adduction began. In previous studies, lift reduction resulted from the exopodite shedding an LEV, which will be discussed in the following section 15. A peak in the lift was observed at approximately $t/T \approx 0.64$, which corresponded with the time when the exopodite began adduction. Lastly, lift decreased when the exopodite reached full adduction during the recovery stroke at $t/T \approx 0.77$.

Drag and thrust production also followed a similar trend as the kinematic data during a beat cycle. For all cupping angles, thrust is largest within the $0.25 < t/T < 0.45$ period. This corresponded to when γ had fully abducted and had yet to initiate the recovery stroke. Maximum drag occurred when the exopodite was fully adducted ($t/T \approx 0.77$). These results reflect the dynamic relationship between the drag, lift, and surface area of beating pleopods, as well as the oscillatory kinematics of the setup. As

expected from Eqs. 2.5 and 2.6, the phases of acceleration and deceleration prescribed by the kinematics align with phases of increased and decreased lift production. In particular, despite a high effective surface area at $t/T = 0.5$, there is no thrust or drag production at this time because it is when the model changes direction.

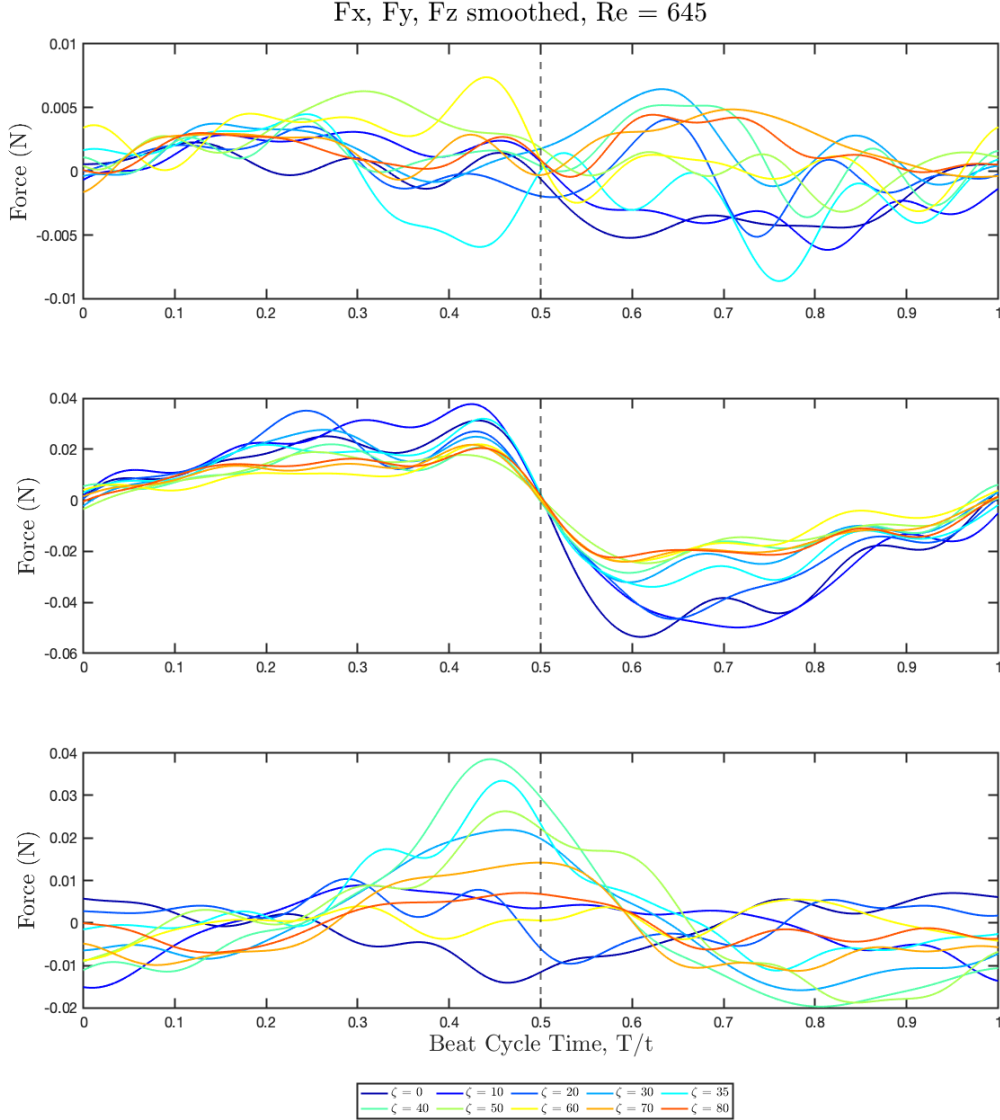


Figure 3.3: **Force measurements along force transducer axes, F_x, F_y, F_z , for all cupping angles at $Re = 645$.** Forces were averaged over four consecutive beat cycles. F_x is the force parallel to the motion of the endopodite throughout the cycle, F_y is the force perpendicular to the endopodite throughout the cycle, and F_z is the force lateral to the endopodite throughout the cycle. F_x and F_y are used to calculate lift and drag. The dashed vertical line represents the transition from the power to the recovery strokes.

In addition to exploring how lift and drag forces evolved during a beat, I quantified how the cupping angle impacted force generation. I focused on the lift and drag-based thrust production during the power stroke because this portion of the cycle remains

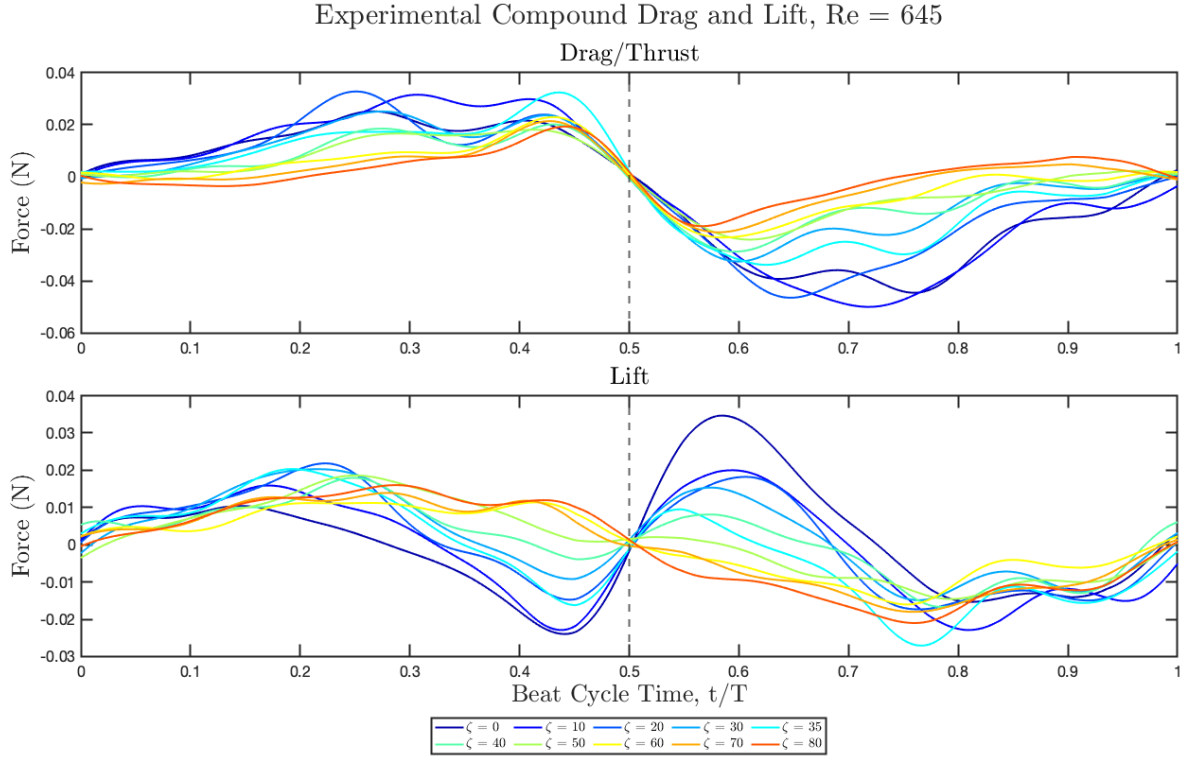


Figure 3.4: **Net lift and drag for all cupping angles at $Re = 645$.** Force measurements along the force transducer axes (F_x , F_y , and F_z) are translated using Eqs. 2.3 and 2.4 to forces into the direction of lift and thrust based on the forward and vertical swimming direction in the lab frame of reference.

the most biologically accurate under the model’s simplifications. While the robotic pleopod is completely rigid and cannot recreate the natural spanwise bending during the recovery phase, its stiffness during the power stroke accurately models shrimp pleopod kinematics during this phase. A tradeoff between lift and thrust can be modulated by varying the cupping angle. During the power stroke, smaller cupping angles led to comparably high thrust but low lift (Fig. 3.5 (a-b)). For example, at $\zeta = 10^\circ$, the ratio of lift to thrust was 0.105. Conversely, large cupping angles ($\zeta \geq 70^\circ$) induced high lift but low thrust. At $\zeta = 50^\circ$ and 60° the ratio of lift to thrust was nearly 1, meaning that the model generated as much lift as thrust during the power stroke. Recall that the biologically observed mean ζ for steady forward swimming was approximately 35° (for *E. superba*). At $\zeta = 35^\circ$, the lift to drag-based thrust ratio was 0.389, indicating that the cupping angle of shrimp during forward swimming was optimized primarily for thrust production but could still produce significant lift. The need to generate this much lift likely results from the necessity for shrimp to counteract gravity because they are negatively buoyant [21]. By assuming that the resulting forces are all in the form of drag-based thrust at $\zeta = 0^\circ$ we can calculate the theoretical relative reduction in the thrust due to increasing the cupping angle (Fig. 3.5 (c)). At $\zeta = 35^\circ$ the model generated up to 94% of the theoretical maximum thrust produced at $\zeta = 0^\circ$, but could still produce about 55% of the maximum lift achieved at $\zeta = 80^\circ$. Compared to a cupping angle of 0° , the 35° cupping angle yielded a remarkable increase in lift production considering the small reduction in thrust. A linear fit of the results in Fig.

3.5 (c) was produced for both the thrust and lift data. For lift, this linear fit was *fraction of maximum force* = $0.0114\zeta + 0.17546$, with $R^2 = 0.854$. For drag, this best fit was *fraction of maximum force* = $-0.0113\zeta + 1.2077$ with $R^2 = 0.889$. The cupping angle at the intersection point indicates optimal thrust and lift production, assuming the desired outcome is to generate as much lift as thrust simultaneously. However, the data shows an important tradeoff between the two, whereby other cupping angles can still promote near-maximum thrust production while enabling the production of sufficient lift to overcome sinking. This crossover occurred around $\zeta = 45^\circ$. Cupping angles of 45° would thus emerge as a good compromise for both forward and vertical propulsion. Although this value is above the biologically relevant cupping angle, it suggests that shrimp do not optimize their cupping angle to maximize both forces. During forward steady swimming, the force required to overcome gravity might not be as high as that required to overcome body drag during locomotion. Therefore, there might not always be a need to employ the ‘optimal’ lift-to-thrust ratio. Which lift-to-thrust ratio is necessary to sustain stable swimming in shrimp is still unclear. However, the cupping angle offers a simple yet effective strategy to modulate thrust and lift production during a beat without requiring the need for energy-demanding active control methods. This prompts deeper investigations of the cupping angle, such as by inducing passive actuation similar to the γ angle or actively controlling the angle of attack of the exopodite during a beat to determine whether keeping it fixed is sufficient or whether some control is needed to tune the performance of the pleopods.

When considering the lift generated during the power stroke, some negative lift was observed at shallow cupping angles, particularly at 0° . While this may seem counter-intuitive in the context of efficient swimming, one must remember that metachronal swimmers activate several appendages at a phase, such that the performance deficiency of one appendage can potentially be offset by that of the other legs. Additionally, shrimp have high maneuverability and can swim upside-down, especially as they graze at the ocean’s surface [22, 39]. Shrimp continually beat their pleopods as they remain at the surface, generating forces that prevent them from sinking. If the cupping angle is dynamic to adjust for different swimming modes, shrimp may utilize this “negative” lift created at high cupping angles to counteract gravity while upside down.

PIV was conducted synchronously with force measurements to determine the connection between the flow field and the measured forces. PIV results show the presence of edge vortices on the exopodite (Fig. 3.6). Vortices formed along both edges of the exopodite at low cupping angles, such as 0° , 10° , and 20° at $t/T = 0.125$ (Figs. 3.7 and 3.8). However, as the cupping angle increased, the trailing edge vortex (TEV) along the exopodite separated and eventually shed at $t/T = 0.125$, enabling the LEV to dominate the flow over the exopodite. As opposed to having vortices distributed along both edges – which in most cases induced a turbulent wake (Fig. 3.7 (a)-(f)) – an attached vortex along the leading edge of the exopodite spread over the exopodite surface can do useful work on the model. In this case, as seen with the force measurements, the LEV imposes work to generate lift. A qualitative analysis of the vorticity fields shows variations in the duration of LEV attachment for different cupping angles (Figs. 3.7 and 3.8). Three timesteps within the power stroke were investigated: $t/T = 0.125$, $t/T = 0.25$, and $t/T = 0.375$. Note that for $\zeta = 60^\circ$, 70° , and 80° , the exopodite does not intersect the laser plane at $t/T = 0.125$. This is due to the reduced γ abduction speed at these high cupping angles.

Overall, LEVs remained attached longer at higher cupping angles. This is most evident when comparing the vorticity fields at $t/T = 0.25$ and $t/T = 0.375$ for cupping angles of 20° (Fig. 3.7 (h),(i)) and 30° (Fig. 3.7 (k),(l)). The LEV remains attached for both cupping angles at $t/T = 0.25$. There are also TEVs visible for both cupping angles at this time. At $t/T = 0.375$, the LEV eventually detached from the exopodite for $\zeta = 20^\circ$, but still remains attached for $\zeta = 30^\circ$. The TEV at $t/T = 0.375$ for $\zeta = 30^\circ$ is also completely separated, promoting a more coherent flow over the exopodite. The prolonged attachment durations of LEVs at higher cupping angles resulted in stronger forces, with higher lift being produced at higher cupping angles (Fig. 3.5).

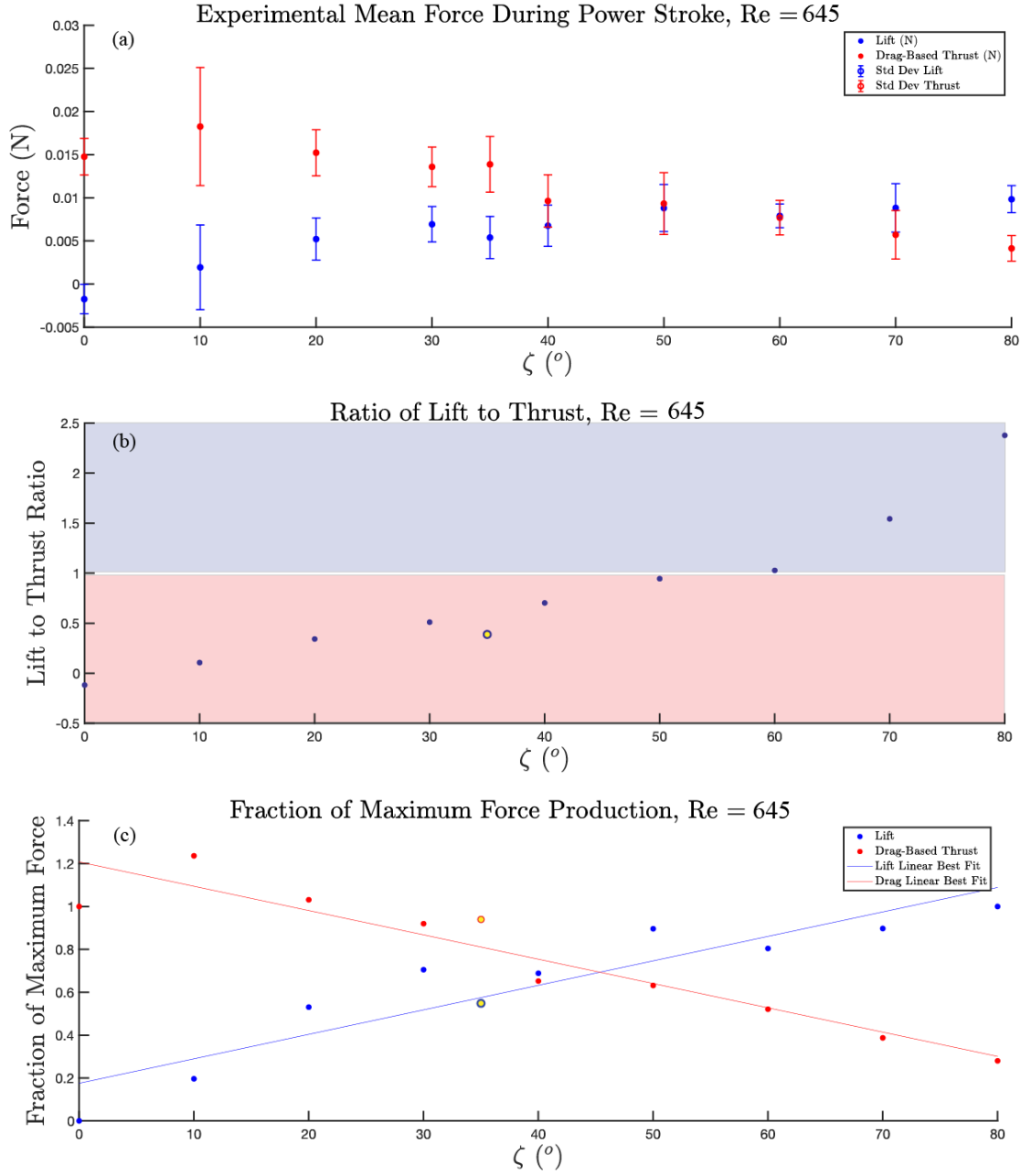


Figure 3.5: **Mean drag and lift produced for all cupping angles during the power stroke.** (a) Mean lift and drag-based thrust forces for all tested cupping angles during the power stroke at $Re = 645$. Lift increases with increasing cupping angle, and drag-based thrust decreases with increasing cupping angle. (b) Lift-to-drag ratio for each cupping angle. Regions of higher lift relative to drag are highlighted by the blue background shading, while the opposite case is highlighted in red. The biologically relevant cupping angle of 35° is marked with yellow. (c) Fraction of maximum force production for each cupping angle. To calculate this fraction, the resulting forces are assumed to come from drag-based thrust exclusively at $\zeta = 0^\circ$. Higher cupping angles are then expressed as a fraction of the drag at $\zeta = 0^\circ$. For lift, the fractions are relative to the maximum observed lift, which occurs at $\zeta = 80^\circ$. Linear best-fit lines for lift (Fraction of lift = $0.0114\zeta + 0.17546$, $R^2 = 0.854$; Fraction of drag = $-0.0113\zeta + 1.2077$, $R^2 = 0.889$) and thrust have been plotted to determine the intersection point at which these ratios are optimized. Points at the biologically relevant cupping angle of 35° are marked with yellow.

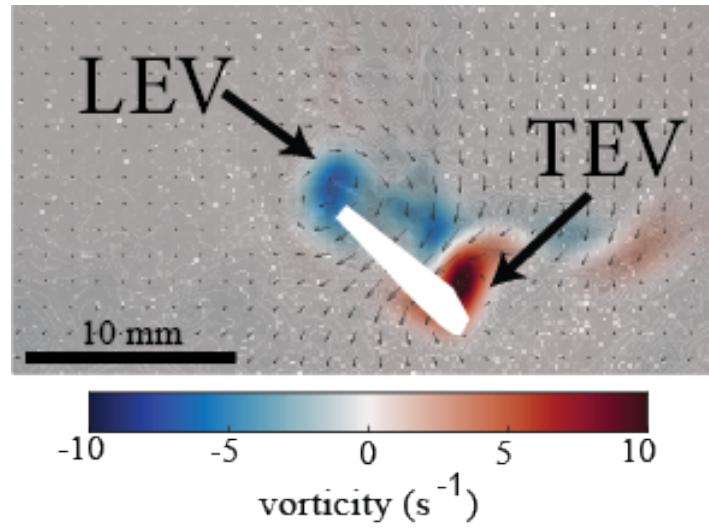


Figure 3.6: **A characteristic leading edge vortex (LEV) and trailing edge vortex (TEV) forming along the exopodite during the power stroke.** This frame is captured at $\zeta = 35^\circ$ at $t/T = 0.25$. The distribution of the LEV over the face of the exopodite is an important factor in improving the effectiveness of the vortex in lift production. In most cases, the TEV separates from the exopodite throughout the stroke.

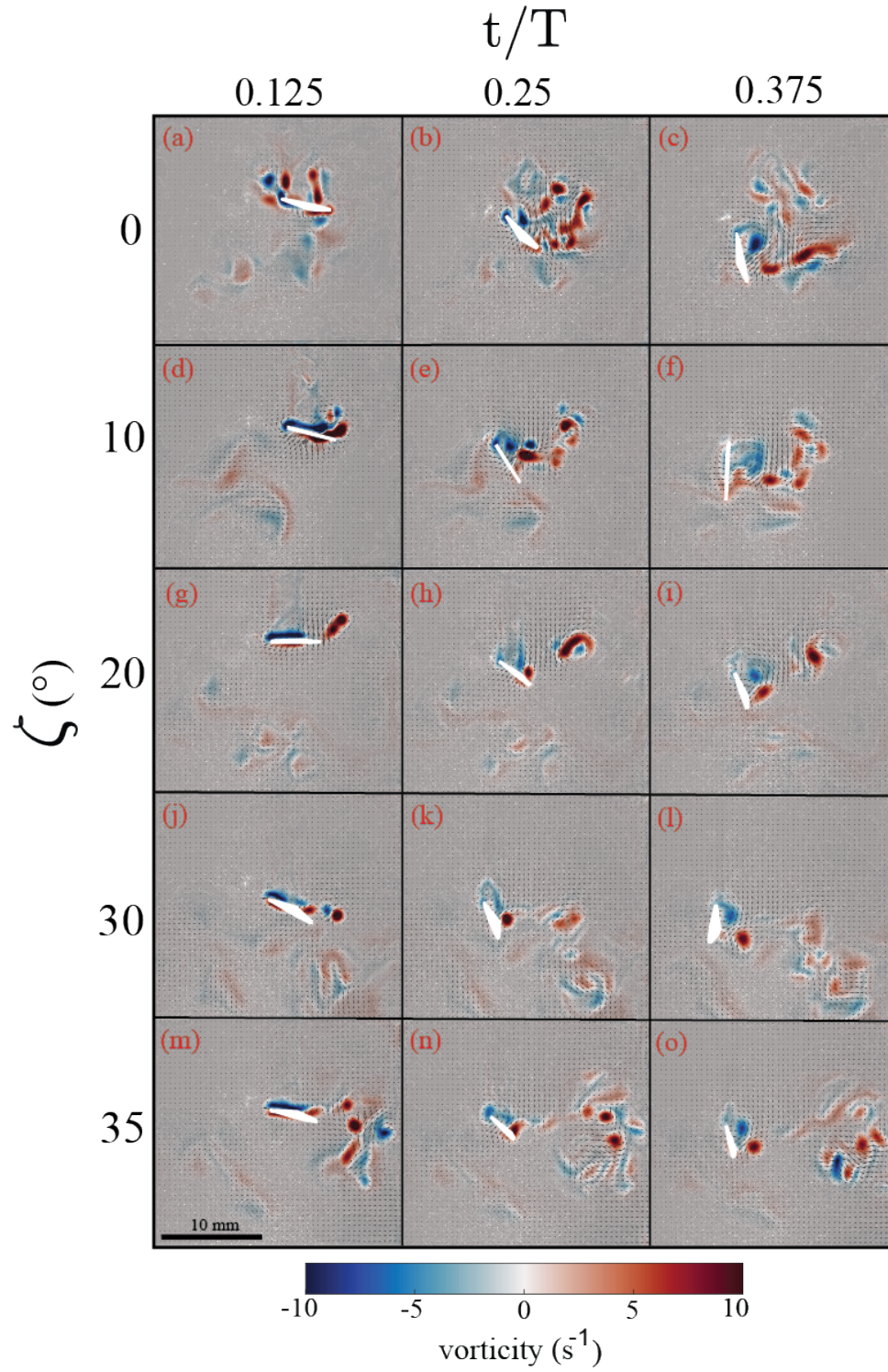


Figure 3.7: **Vorticity fields for $\zeta = 0^\circ$ - 35° at $t/T = 0.125$, 0.25 , and 0.375 for $\text{Re} = 968$** The exopodite cross-section is masked in white.

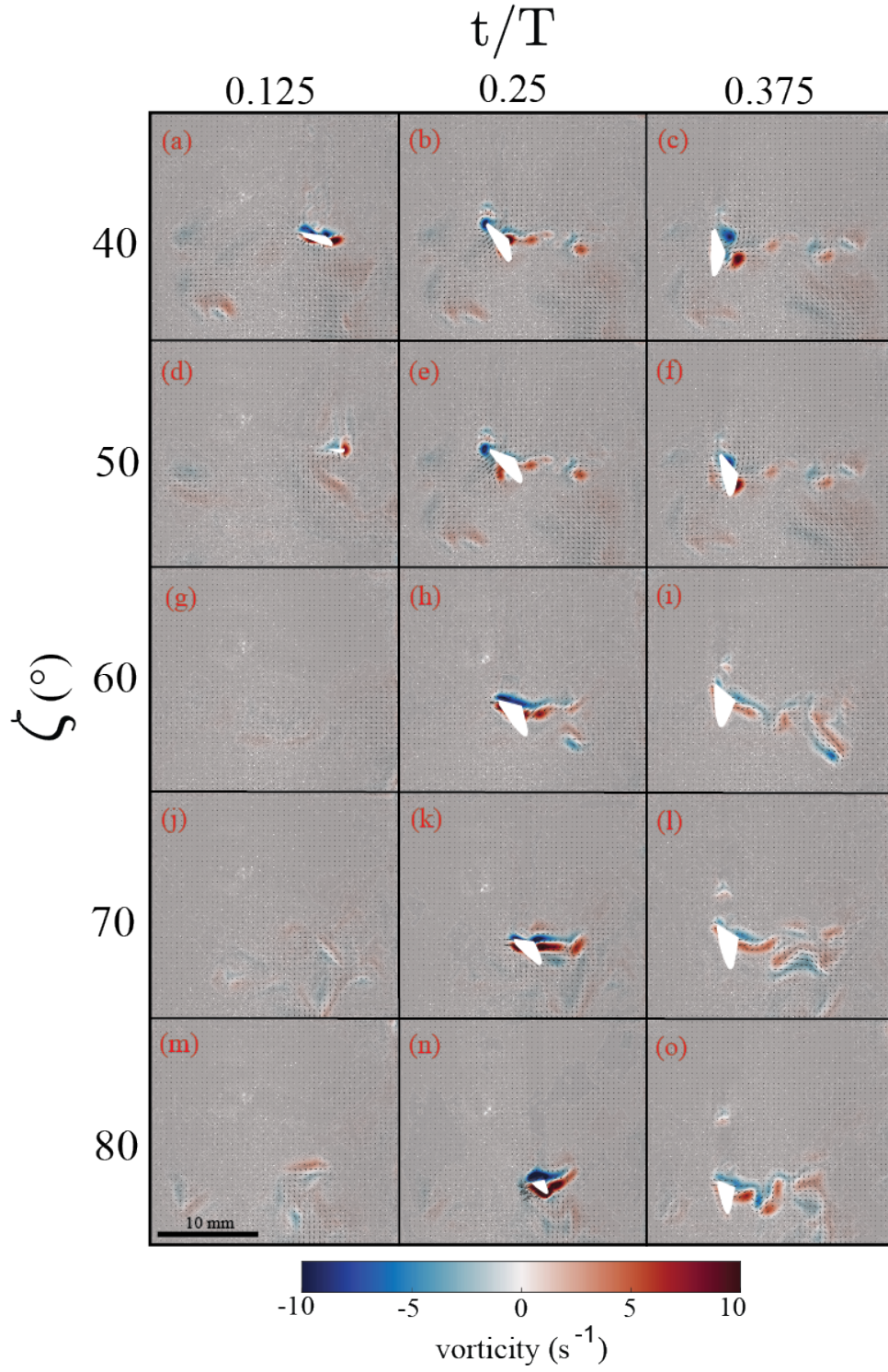


Figure 3.8: **Vorticity fields for $\zeta = 40^\circ$ - 80° at $t/T = 0.125$, 0.25 , and 0.375 for $\text{Re} = 968$** The exopodite cross-section is masked in white. The exopodite is not visible at $t/T = 0.125$ for $\zeta = 60^\circ$, 70° , and 80° because the exopodite abducted slower and with a lesser amplitude compared with shallower cupping angles.

Chapter 4

Conclusion and Outlook

This thesis investigated the role of the cupping angle found in shrimp pleopods on lift and thrust generation during swimming. I designed and 3D printed a scaled robotic pleopod analog to comprehensively investigate the effects of the cupping angle beyond what is achievable through biological observations alone. I found that the cupping angle dictates how fast the exopodite can abduct and adduct during the beat cycle to generate useful forces longer during a beat. The results show that cupping angles of 40° result in the fastest abduction and adduction of the exopodite. Other moderate cupping angles of 20° , 30° , and the biologically relevant cupping angle of 35° also contribute to fast γ modulation and, consequentially, fast surface area modulation for thrust production and drag reduction during the power and recovery strokes, respectively. Varying the cupping angle yields a tradeoff between the lift and thrust produced during the power stroke. I found that a cupping angle around 45° is optimal for minimizing the tradeoff between thrust and lift such that these forces are nearly equal. Interestingly, the natural cupping angle found in shrimp is different and is only about 35° . The results suggest that shrimp may leverage this tradeoff between lift and thrust to generate near-maximum maximum thrust while still producing about 55% of the theoretical maximum lift to overcome sinking under their own weight. While I used a fixed cupping angle, it is not excluded that this angle be modulated during swimming, either passively or actively, to optimize force production and the required balance between thrust and lift needed to sustain other swimming behaviors. Relatively extreme cupping angles, such as 0° or 80° , may be used in unusual swimming modes, such as backward swimming or upside-down swimming, when only thrust or lift is required. Future robotic studies could examine how a dynamic cupping angle affects force production in the context of optimization for steady swimming and maneuverability. Additionally, considerations should be given to untethering the robot to obtain force estimates under natural, variable free-stream velocities. The performance of this shrimp-inspired model illuminates its potential for integration into ocean exploration and monitoring devices. Although it is unclear to what extent shrimp actively abduct and adduct their exopodite, the entirely passive implementation of this motion in the model makes it valuable for simplified yet effective engineering solutions. This approach allows for emulating power and recovery stroke cycles without additional motors, thereby simplifying actuation and minimizing power requirements.

Chapter 5

Appendix

5.1 Model Design and Experimental Setup

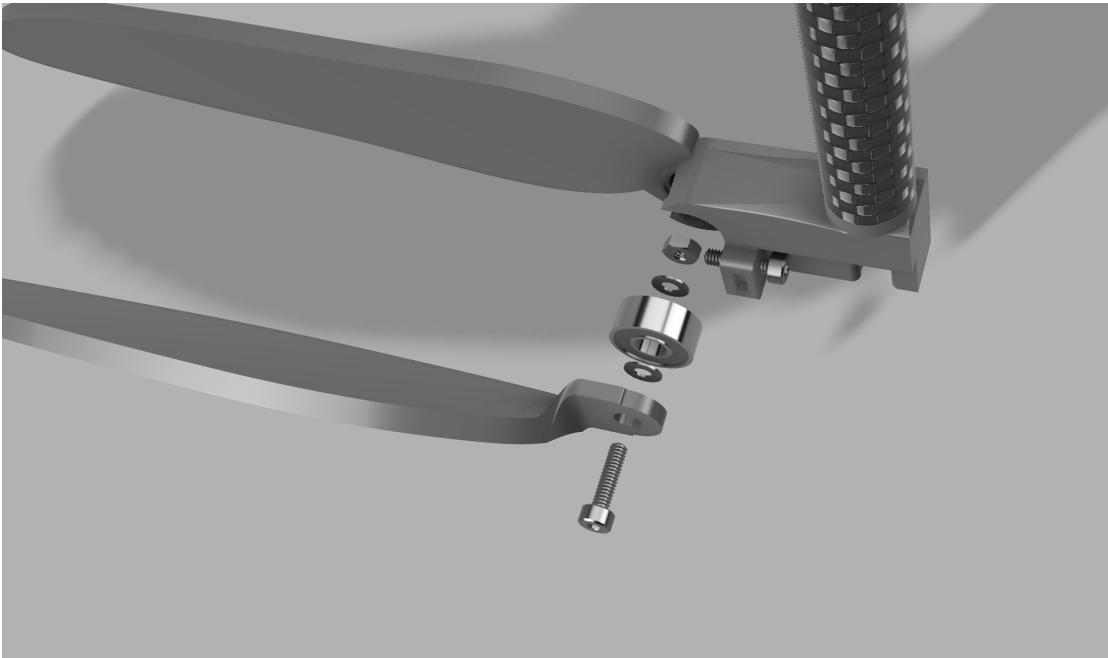


Figure 5.1: **Exploded CAD model of the ramal joint.** The ball bearing mitigates friction when the exopodite rotates.



Figure 5.2: **3D-printed swappable proximal sections of the exopodite with varying cupping angle.** Asymmetric placement of the dovetail ensures that the dovetail cannot be inserted backward into the exopodite.

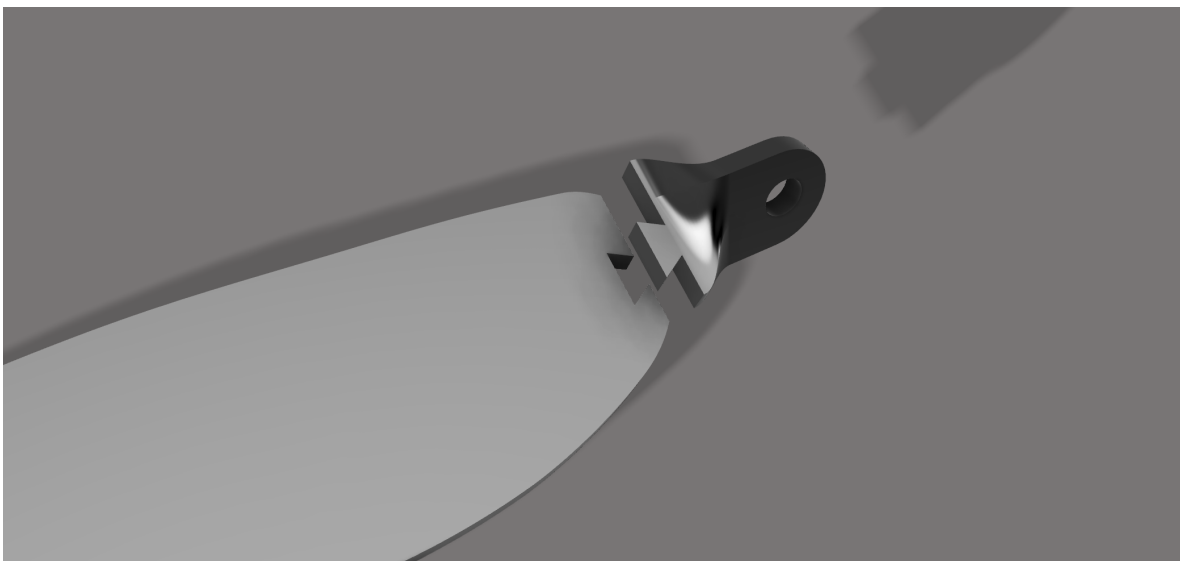


Figure 5.3: **Cupping angle adjustment method.** A dovetail is used to connect the main body of the exopodite to the swappable proximal section responsible for setting the cupping angle. This modular design ensured that only the proximal section needed to be swapped, thus keeping the same exopodite (with its specific neutrally buoyant properties) for all the trials. Printing the models using resin produced a tight-fitting dovetail that required no additional fasteners. This design did not compromise the overall stiffness of the exopodite.

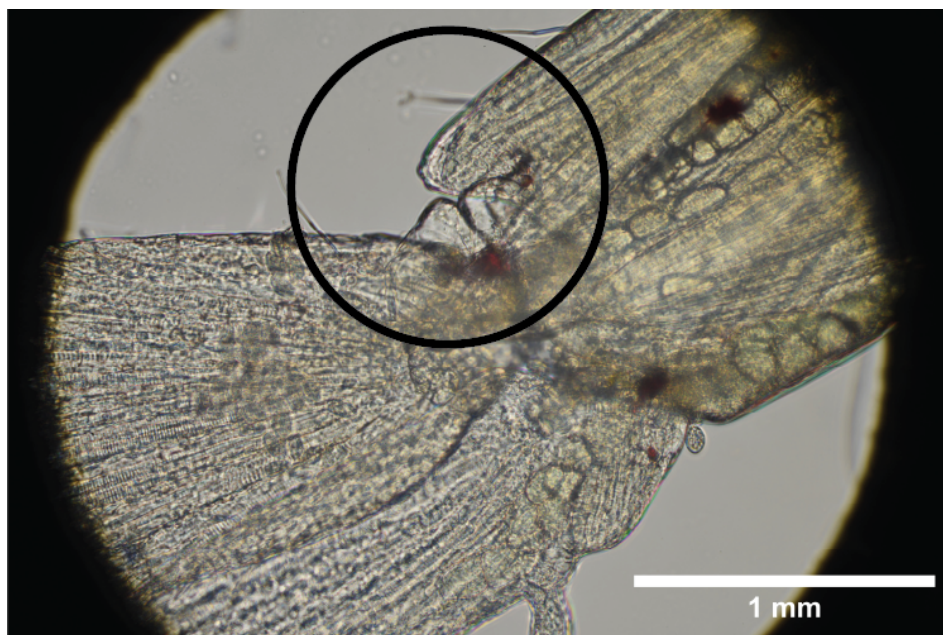


Figure 5.4: γ -limiting joint structure in Hawaiian red shrimp, *Halocaridina rubra*. This joint structure inspired the addition of a bolt in the robotic pleopod to prevent the robotic exopodite from abducting past $\gamma = 70^\circ$.

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